

Bernard Birendra Das

Machine Learning Engineer

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PROFILE SUMMARY

Machine Learning Engineer and PhD Scholar in Data Science with 2+ years of professional software engineering experience designing, training, and deploying production ML systems. Specialized in computer vision, deep learning backbones, inference acceleration (TensorRT, ONNX, Triton), and distributed pipelines. Proven success in shipping high-accuracy models with optimized latency for real-time video processing and diagnostic applications.

WORK EXPERIENCE

Simplify3x Software Private Limited • Software Engineer

January 2023 - May 2025 | 2 years 5 months | Bengaluru, Karnataka

- Developed and maintained two impactful modules of SimplifyQA product for the new UI, namely PDF Reports and Pipeline Executions.
- Designed and implemented RESTful APIs and microservices that led to significant improvements in response times and improving overall system scalability.
- Developed dynamic, data-driven dashboards using Angular and D3.js, enabling real-time visualization of key performance metrics for business stakeholders.

PROJECTS

Real-Time Computer Vision & Landmark System • MediaPipe, OpenCV, Streamlit, WebRTC, Python

2026

- Engineered a real-time multi-task vision pipeline using MediaPipe Tasks API and OpenCV for hand tracking and pose estimation.
- Built low-latency video streaming using WebRTC and deployed interactive web controls for workout analytics.

Brain Tumor Segmentation & Classification • TensorFlow, Keras, VGG16, Flask, NumPy

2026

- Fine-tuned a deep VGG16 convolutional network for 3-class MRI tumor diagnosis, achieving 99.36% test accuracy.
- Containerized model serving with Flask REST API and optimized inference pipelines for minimal response latency.

EDUCATION

CHRIST (Deemed-to-be-University), Bangalore

PhD in Data Science - 3.6/4.0 GPA

Dec 2024 - Present

MCA with Data Analytics - 8.66 CGPA

June 2021 - June 2023

St. Xavier's College (Autonomous), Kolkata

BSc (Honours) Computer Science - 7.20 CGPA

June 2018 - June 2021

Auxilium Convent School, Barasat

12th Grade, ISC Board - 71.20%

June 2016 - June 2018

10th Grade, ICSE Board - 76.40%

June 2004 - June 2016

SKILLS

PROGRAMMING

Languages:

Python • C++ • Java • SQL • R

Machine Learning & DL:

PyTorch • TensorFlow • Hugging Face • Scikit-learn • OpenCV • JAX

Inference Acceleration:

TensorRT • ONNX Runtime • Triton • vLLM • Quantization

SYSTEMS & PLATFORMS

Data & Distributed Systems:

Apache Spark • Apache Kafka • BigQuery

MLOps & Cloud Platforms:

Kubernetes • Docker • MLflow • AWS SageMaker • GCP Vertex AI

CORE COMPETENCIES

Computer Vision • Model Optimization • Distributed Training • Low-Latency Serving • Transfer Learning

CERTIFICATIONS

Foundational C# with Microsoft

Developer Certification

Sept 2023

AWS Academy Graduate

AWS Academy Cloud Foundations Nov 2022